1 Squeeze Theorem

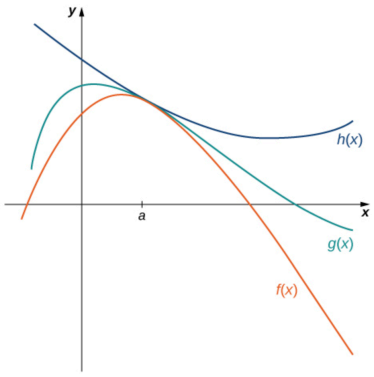
Theorem 1.1 (Squeeze Theorem)**.** *Let f, g, and h be defined for all $x \neq a$ over an open interval containing a. If*

$$f(x) \leq g(x) \leq h(x)$$

for all $x \neq a$ in an open interval containing a and

$$\lim_{x \to a} f(x) = L = \lim_{x \to a} h(x)$$

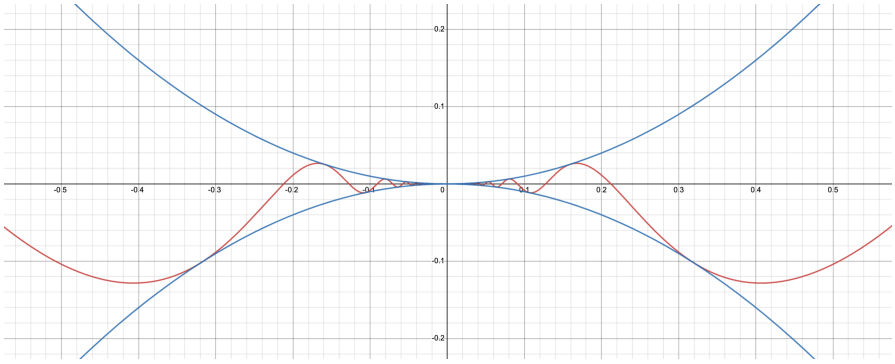
where $L \in \mathbb{R}$, then $\lim_{x \to a} g(x) = L$.



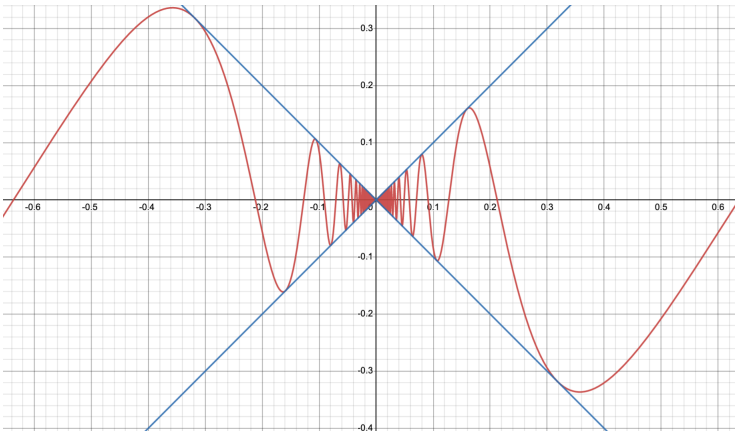
Example 1.2. We have proved in the notes that $|\sin(\theta)| \leq |\theta| \iff -|\theta| \leq \sin(\theta) \leq |\theta|$ for all $\theta \in \mathbb{R}$. Since $\lim_{\theta \to 0} |\theta| = 0 = \lim_{\theta \to 0}(-|\theta|)$, the Squeeze Theorem implies $\lim_{\theta \to 0} \sin(\theta) = 0$.

1. Prove the Squeeze Theorem.

1. Use the Squeeze Theorem to find $\lim_{x\to 0} x^2 \cos(\frac{1}{x})$.

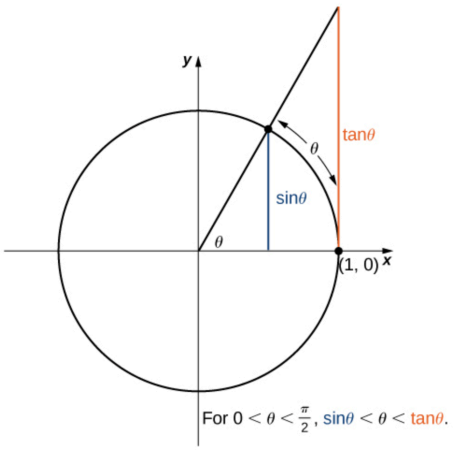


2. Use the Squeeze Theorem to find $\lim_{x\to 0} x\cos(\frac{1}{x})$.



2 Two Important Trig Limits

3. Consider the limit $\lim_{\theta \to 0} \frac{\sin(\theta)}{\theta}$.



(a) Using the figure above, convince yourself that $0 < \theta < \frac{\pi}{2}$, $\sin(\theta) < \theta < \tan(\theta)$.

(b) Show that the above implies $\cos(\theta) < \frac{\sin(\theta)}{\theta} < 1$ for $0 < \theta < \frac{\pi}{2}$.

(c) Substitute $\theta \mapsto -\theta$ in the inequality above to show that the same inequality holds for $-\frac{\pi}{2} < \theta < 0$. Thus, the inequality holds on $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

(d) Compute $\lim_{\theta \to 0} \frac{\sin(\theta)}{\theta}$.

4. Find $\lim_{\theta \to 0} \frac{1-\cos(\theta)}{\theta}$. (*Hint*: Multiply by the conjugate.)

5. Find $\lim_{\theta \to 0} \frac{1-\cos(\theta)}{\sin(\theta)}$.

3 More special trig limits

In the previous subsection we used the Squeeze Theorem to prove the following special trig limits:

$$\lim_{\theta\to 0}\frac{\sin(\theta)}{\theta}=1, \tag{3.1}$$

$$\lim_{\theta\to 0}\frac{1-\cos(\theta)}{\theta}=0. \tag{3.2}$$

From these, we can prove many more.

Example 3.3. Consider $\lim_{x\to 0}\frac{\sin(6x)}{x}$. This looks similar to (3.1), except that the argument of the sine function does not match the denominator. We can make these match if we multiply the numerator and denominator by 6:

$$\begin{aligned}\lim_{x\to 0}\frac{\sin(6x)}{x}&=\lim_{x\to 0}\left(\frac{6}{6}\cdot\frac{\sin(6x)}{x}\right)\\&=\lim_{x\to 0}\frac{6\sin(6x)}{6x}\\&=6\lim_{x\to 0}\frac{\sin(6x)}{6x}\qquad\text{(by the Constant Multiple Law).}\end{aligned}$$

Now, this looks a lot like (3.1) with $\theta=6x$, but making this substitution gives $\lim_{\frac{\theta}{6}\to 0}\frac{\sin(\theta)}{\theta}$, which is not quite the same as (3.1) due to the $\frac{\theta}{6}$ under the limit sign. However, as $\theta\to 0$, we should still have $\frac{\theta}{6}\to 0$, so we are tempted to conclude that

$$\lim_{x\to 0}\frac{\sin(6x)}{6x}=\lim_{\frac{\theta}{6}\to 0}\frac{\sin(\theta)}{\theta}=1. \tag{3.4}$$

Let's prove that this is the case. Let $\varepsilon>0$. Since $\lim_{\theta\to 0}\frac{\sin(\theta)}{\theta}=1$, there exists a corresponding $\delta>0$ such that $0<|\theta|<\delta$ implies $|\frac{\sin(\theta)}{\theta}|<\varepsilon$. Choose $\delta'=\frac{\theta}{6}$. It follows that

$$0<\left|\frac{\theta}{6}\right|<\delta' \implies 0<|\theta|<\delta \implies \left|\frac{\sin(\theta)}{\theta}-1\right|<\varepsilon.$$

This proves (3.4).

In a similar way, one can prove the following:

Proposition 3.5. *If* $\lim_{x\to 0}f(x)=L$ *and* $a\neq 0$*, then* $\lim_{x\to 0}f(ax)=L$.

Instructions: Follow the strategy in the previous example to calculate the following trig limits.

1. $\lim_{x\to 0} \dfrac{1-\cos(3x)}{x}$

2. $\lim_{x\to 0} \dfrac{\sin(x)}{5x}$

3. $\lim_{\theta\to 0} \dfrac{\theta}{\sin(4\theta)}$

4. $\lim_{x\to 0} \dfrac{\sin(6x)}{\sin(2x)}$

5. $\lim_{x\to 0} \dfrac{\tan(3x)}{2x}$

6. $\lim_{x\to 0} \dfrac{\sin(5x)}{2x}$

7. $\lim_{x\to 0} \dfrac{\sin^2(2x)}{x^2}$

8. $\lim_{\theta\to 0} \dfrac{\tan(4\theta)}{\sin(6\theta)}$

9. $\lim_{x\to 0} \dfrac{x}{x+\sin(x)}$

10. $\lim_{x\to 0} \dfrac{\tan(x)}{x}$

11. $\lim_{x\to \infty} \dfrac{x+\sin(x)}{x+\cos(x)}$

12. $\lim_{x\to \infty} x\sin\left(\dfrac{1}{x}\right)$